Data first

AI · Level 3 · Lesson 1



Today we will

- Plan a dataset: what to collect, the labels, how many and how varied.
- Spot wrong labels, too few examples and one kind far outnumbering another.
- Collect data fairly: ask permission and keep personal details out.

Remember

- A model only knows its examples: your data is its only textbook.
- Check for wrong labels, too few examples and unbalanced labels. Variety matters as much as numbers.
- Data comes from people: ask first, say what it’s for, and leave names out.

Data detectives

1. Pick (3 min): a sorting job with two labels.
2. Plan (5 min): what, labels, how many, how varied.
3. Collect (8 min): 12 labelled examples, one label wrong on purpose.
4. Swap (5 min): a partner hunts for problems.
5. Fix (4 min): three things to add or change.

Works in a group of three or four, or at home with a family member. Use drawings if there is no camera, and never photograph people without asking.

Let’s discuss

1. Why can’t a model notice that one of its labels is wrong?
2. A dataset has 90 healthy leaves and 10 sick ones. What could go wrong?
3. Whose permission would you need to collect voices at school?

Before you go

- One thing my dataset would be missing, and how I would collect it fairly.